Window-covering words and a bouquet of cycles

Exact enumeration of the OEIS family A242239

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Abstract

For integers $k, n \geq 1$, let $T(n,k)$ be the number of canonical words of length $n+k+1$ over $\{0,1,\ldots,k\}$ such that every consecutive block of length $k+2$ contains every symbol. This is the table OEIS A242239. We prove that these words, modulo relabeling, are encoded by length-n walks on a directed bouquet whose cycle lengths are exactly $1,2,\ldots,k+1$. Consequently,

$$\sum_{n\geq 1} T(n,k)x^n = \frac{\displaystyle\sum_{j=1}^{k+1} \frac{(k+2-j)(kj+2)}{2} x^j}{1-x-x^2-\cdots-x^{k+1}}.$$

This proves the empirical $(k+1)$-bonacci recurrence recorded for every column of A242239, with exact starting index $n = k+2$, and also proves the stable-row formula

$$T(n,k) = \frac{2^n-1}{2}k^2 + \frac{2n+1}{2}k + 1 \qquad (k \geq n-3).$$

For $k = 4$ the result proves the empirical recurrence and generating function of A242235. Using the already known generating function of A145113, it also proves Mathar's conjectured identity for that sequence.

1 Introduction and main result

Set $\Sigma_k = \{0,1,\ldots,k\}$, so $|\Sigma_k| = k+1$. A word over Σ_k is called *canonical* if the first occurrences of its distinct symbols appear in the order $0,1,2,\ldots$; equivalently, it is a restricted growth word. The table A242239 is defined by

$$T(n,k) = \#\left\{ w \in \Sigma_k^{n+k+1} : \begin{array}{l} w \text{ is canonical, and} \\ \text{every } (k+2)\text{-factor of } w \text{ contains } \Sigma_k \end{array} \right\}. \tag{1}$$

The OEIS entry records, as empirical observations, both a $(k+1)$-step Fibonacci recurrence for each column and a quadratic formula for each row once k is sufficiently large [1]. The special column $k = 4$ is A242235; its recurrence and generating function are likewise marked empirical, and an identity proposed by Mathar is marked conjectural [2].

The main result gives the complete ordinary generating function in closed form.

Theorem 1.1 (Exact generating function)**.** *For every $k \geq 1$, define*

$$D_k(x) = 1 - x - x^2 - \cdots - x^{k+1}$$

and

$$N_k(x) = \sum_{j=1}^{k+1} c_{k,j} x^j, \qquad c_{k,j} = \frac{(k+2-j)(kj+2)}{2}.$$

Then

$$\mathcal{A}_k(x) := \sum_{n\geq 1} T(n,k)x^n = \frac{N_k(x)}{D_k(x)}. \tag{2}$$

The coefficients $c_{k,j}$ are positive integers. The integrality is immediate: if k or j is even then $kj+2$ is even, while if both are odd then $k+2-j$ is even.

Corollary 1.2 (Column recurrence)**.** *For every $k \geq 1$,*

$$T(n,k) = \sum_{i=1}^{k+1} T(n-i,k) \qquad (n \geq k+2).$$

This range is exact: with the convention $T(m,k) = 0$ for $m \leq 0$, the same relation fails for every $1 \leq n \leq k+1$, because its residual is the positive coefficient $c_{k,n}$.

Corollary 1.3 (Stable rows)**.** *For $n, k \geq 1$ with $k \geq n-3$,*

$$T(n,k) = \frac{2^n-1}{2}k^2 + \frac{2n+1}{2}k + 1. \tag{3}$$

As of the access date in the bibliography, these formulas remain labelled empirical on A242239 and its relevant column entries. We are not aware of a previous proof; this statement is a qualified literature-status report, not a claim that an exhaustive priority search is possible.

2 From words to a defect-pair automaton

Put $q = k+1$. A block of length q is *rainbow* if all its letters are distinct. Otherwise, a valid block will have one repeated symbol. If its two copies occur in positions $1 \leq i < j \leq q$, call its type (i,j).

Lemma 2.1 (Relabeling orbits)**.** *Let $N \geq q+1$, and let $\mathcal{W}_k(N)$ be the set of labelled words of length N over Σ_k satisfying the sliding-window condition in* (1)*. The symmetric group S_q acts freely on $\mathcal{W}_k(N)$ by relabeling, and every orbit contains exactly one canonical word. Hence*

$$T(n,k) = \frac{|\mathcal{W}_k(n+q)|}{q!}.$$

Proof. The first window of length $q+1$ contains every symbol, so every word uses all q symbols. A permutation fixing the word therefore fixes every symbol and is the identity. In each orbit, relabel the symbols by their first-occurrence order; this produces exactly one canonical representative. □

Lemma 2.2 (Block classification)**.** *Every length-q block occurring in a word of $\mathcal{W}_k(N)$ contains at least $q-1$ distinct symbols. It is therefore either rainbow or has a unique type (i,j) and misses exactly one symbol.*

Proof. Any length-q block extends to a valid length-$(q+1)$ window, to the right unless it is the final block and to the left otherwise. Deleting one letter from a window containing all q symbols leaves at least $q-1$ distinct symbols. A non-rainbow block consequently contains exactly one doubled symbol and misses exactly one symbol. □

Lemma 2.3 (Transitions)**.** *Let $B = b_1 \cdots b_q$ be the final block of a valid prefix.*

(a) *If B is rainbow, every symbol may be appended. Appending b_1 gives another rainbow block, while appending b_{p+1} gives type (p, q) for $1 \le p \le q-1$.*

(b) *If B has type (i, j) and misses m, the only legal appended symbol is m. The new block is rainbow when $i = 1$, and has type $(i-1, j-1)$ when $i > 1$.*

Each listed transition is realized by exactly one appended symbol.

Proof. The newly created window is Bx, and it is valid precisely when the symbols in B together with x cover Σ_k. In the rainbow case, shifting removes b_1 and appending a copy of b_r either replaces the removed symbol ($r = 1$) or creates a doubled symbol in positions $(r-1, q)$. In the defect case, the missing symbol is forced; shifting either removes the first copy of the doubled symbol or shifts both copies one position to the left. □

The quotient graph has a rainbow state R and one state (i, j) for each $1 \le i < j \le q$. The transitions in lemma 2.3 reveal a simpler structure. For each $\ell = 2, \ldots, q$, the states

$$(\ell - 1, q), (\ell - 2, q - 1), \ldots, (1, q - \ell + 2)$$

form, together with R, a directed cycle of length ℓ. Different values of ℓ give disjoint non-hub vertices, and R also has a self-loop. Thus the quotient graph is the directed bouquet

$$\mathcal{B}_q = C_1 \vee C_2 \vee \cdots \vee C_q,$$

where the cycles share only the hub R.

Proposition 2.4 (Bouquet encoding)**.** *For every $n, k \ge 1$, $T(n, k)$ is the number of directed walks of length n in $\mathcal{B}_{k+1}$, with every vertex allowed as a starting vertex and every vertex allowed as an ending vertex.*

Proof. Choose the initial block of length q and record its type. By lemma 2.3, a state walk determines each subsequently appended letter uniquely, and every state edge is realizable. Hence labelled words are in bijection with pairs consisting of a concrete initial block and a state walk of length n.

For each state type, the set of concrete blocks has exactly one S_q-orbit. There are $q!$ rainbow blocks. For a fixed defect type (i, j), choose the doubled symbol and then injectively assign the remaining $q - 2$ positions from the other $q - 1$ symbols; again there are $q!$ blocks. Relabeling changes the concrete block but not the state walk. Applying lemma 2.1 therefore leaves one initial orbit for every vertex of $\mathcal{B}_q$, proving the result. □

3 Enumeration of walks on the bouquet

Let

$$H_k(x) = 1 + \sum_{j=1}^{k} (k + 1 - j)x^j, \qquad s_k = 1 + \frac{k(k+1)}{2} = \frac{k^2 + k + 2}{2}. \tag{4}$$

The number s_k is the number of vertices of $\mathcal{B}_{k+1}$.

Let $P_k(x)$ be the generating function for walks that start at the hub R and may end anywhere. Such a walk consists uniquely of a sequence of complete returns to the hub,

followed by a proper prefix of one cycle. The complete return lengths are $1, 2, \ldots, k+1$, one of each length. A proper prefix of length $j \geq 1$ can be chosen in $k+1-j$ ways. Therefore

$$P_k(x) = \frac{H_k(x)}{D_k(x)}. \tag{5}$$

A vertex at forced distance i from the hub contributes

$$1 + x + \cdots + x^{i-1} + x^i P_k(x),$$

and there are $k+1-i$ such vertices. Summing over all vertices, including the hub, gives the generating function for walks of all lengths:

$$\sum_{n \geq 0} T_0(n,k) x^n = H_k(x) P_k(x) + G_k(x), \tag{6}$$

where $T_0(0,k) = s_k$, $T_0(n,k) = T(n,k)$ for $n \geq 1$, and

$$G_k(x) = \sum_{i=1}^{k} (k+1-i)(1 + x + \cdots + x^{i-1}) = \frac{s_k - H_k(x)}{1-x}.$$

The elementary identity

$$(1-x)H_k(x) = D_k(x) + kx \tag{7}$$

now removes the only apparent extra denominator. From (5)–(6),

$$\begin{aligned} \mathcal{A}_k(x) &= \frac{H_k(x)^2}{D_k(x)} + \frac{s_k - H_k(x)}{1-x} - s_k \\ &= \frac{H_k(x)^2}{D_k(x)} + \frac{s_k x - H_k(x)}{1-x}. \end{aligned}$$

Consequently,

$$\begin{aligned} D_k(x)\mathcal{A}_k(x) &= \frac{(1-x)H_k(x)^2 + D_k(x)(s_k x - H_k(x))}{1-x} \\ &= \frac{x\big(kH_k(x) + s_k D_k(x)\big)}{1-x}. \end{aligned} \tag{8}$$

The polynomial inside the last quotient vanishes at $x = 1$, since $H_k(1) = s_k$ and $D_k(1) = -k$. Thus the division by $1-x$ is exact.

For completeness, let $N_k(x) = \sum_{j=1}^{k+1} c_{k,j} x^j$. Dividing (8) by taking partial sums of coefficients gives

$$\begin{aligned} c_{k,j} &= k + s_k + \sum_{r=1}^{j-1} \big(k(k+1-r) - s_k\big) \\ &= \frac{(k+2-j)(kj+2)}{2}, \qquad 1 \leq j \leq k+1. \end{aligned}$$

This proves theorem 1.1, and corollary 1.2 follows by coefficient comparison in $D_k(x)\mathcal{A}_k(x) = N_k(x)$.

Table 1: The first numerator polynomials, generated directly from $c_{k,j} = ((k+2-j)(kj+2))/2$.

k	$N_k(x)$
1	$3x + 2x^2$
2	$6x + 6x^2 + 4x^3$
3	$10x + 12x^2 + 11x^3 + 7x^4$
4	$15x + 20x^2 + 21x^3 + 18x^4 + 11x^5$
5	$21x + 30x^2 + 34x^3 + 33x^4 + 27x^5 + 16x^6$
6	$28x + 42x^2 + 50x^3 + 52x^4 + 48x^5 + 38x^6 + 22x^7$
7	$36x + 56x^2 + 69x^3 + 75x^4 + 74x^5 + 66x^6 + 51x^7 + 29x^8$
8	$45x + 72x^2 + 91x^3 + 102x^4 + 105x^5 + 100x^6 + 87x^7 + 66x^8 + 37x^9$

4 The stable-row formula

We prove corollary 1.3 directly from the bouquet. Let

$$r_m = [x^m]P_k(x)$$

be the number of length-m walks starting at the hub. From $D_k(x)P_k(x) = H_k(x)$, a short induction gives

$$r_0 = 1, \qquad r_m = k2^{m-1} + 1 \quad (1 \le m \le k+2), \tag{9}$$

and at the next index

$$r_{k+3} = k2^{k+2} - k + 1. \tag{10}$$

Indeed, for $1 \le m \le k$, the recurrence for r_m has forcing term $k + 1 - m$; summing the earlier values in (9) gives the claimed value. The cases $m = k+1, k+2, k+3$ then follow from the homogeneous recurrence with the appropriate $k+1$ preceding terms.

A vertex at distance i from the hub has exactly one length-n continuation if $n \le i$, and otherwise reaches the hub after i steps and has r_{n-i} continuations. Therefore

$$T(n,k) = r_n + \sum_{i=1}^{k}(k+1-i)\begin{cases}1, & n \le i,\\ r_{n-i}, & n > i.\end{cases} \tag{11}$$

If $n \le k$, substituting (9) into (11) gives

$$\begin{aligned} T(n,k) &= k2^{n-1} + 1 + \sum_{i=1}^{n-1}(k+1-i)(k2^{n-i-1}+1) + \sum_{i=n}^{k}(k+1-i) \\ &= \frac{2^n-1}{2}k^2 + \frac{2n+1}{2}k + 1. \end{aligned}$$

For $n = k+1$ and $n = k+2$, the same substitution applies with the final sum empty. For $n = k+3$, use (10) for the first term and (9) for every term in the remaining sum. The same simplification results. These four cases are exactly $n \le k+3$, proving corollary 1.3.

5 The column A242235 and Mathar's identity

Taking $k = 4$ in theorem 1.1 gives

$$\sum_{n\geq 1} T(n,4)x^n = \frac{x(15+20x+21x^2+18x^3+11x^4)}{1-x-x^2-x^3-x^4-x^5}. \tag{12}$$

Thus A242235 satisfies the pentanacci recurrence from $n = 6$, with initial values

$$15, 35, 71, 139, 271.$$

This proves the empirical formulas recorded on the OEIS entry [2].

Let $b(m)$ be A145113, the number of length-m binary words with fewer than five zeros between consecutive ones. Its generating function is already recorded on that entry as

$$B(x) = \sum_{m\geq 0} b(m)x^m = \frac{1-x+x^6}{1-3x+2x^2+x^6-x^7} = \frac{1-x+x^6}{(1-x)^2 D_4(x)}. \tag{13}$$

See [3]. It can also be obtained from the standard seven-state automaton that records the number of zeros since the most recent one.

Combining (12) and (13) and clearing the common denominator gives the exact identity

$$\mathcal{A}_4(x) = 16xB(x) + \frac{4x}{(1-x)^2} - \frac{5x}{1-x}. \tag{14}$$

Hence

$$T(n,4) = 16b(n-1) + 4n - 5 \qquad (n \geq 1). \tag{15}$$

For $n \geq 2$, this is precisely Mathar's conjecture $T(n,4) = 16\,\mathrm{A145113}(n-1) + \mathrm{A004767}(n-2)$, because $\mathrm{A004767}(m) = 4m+3$ [4]. At $n = 1$, (15) remains valid, but the A004767 notation requires polynomially extending $4m+3$ to $m = -1$; it is not an ordinary in-range OEIS sequence evaluation.

6 Scope, verification, and disclosure

The bouquet reduction depends critically on the window length being exactly one larger than the alphabet size. In that regime, deleting one letter from a covering window leaves either a rainbow block or a block with one doubled and one missing symbol. For longer windows, multiple defects can occur, and the same state space does not apply without further work.

No step of the proof relies on numerical evidence. The accompanying verification program is `verification/verify_hardin_family.py`. It uses only the Python standard library and independently checks:

1. the closed numerator formula against walks on the bouquet;
2. a raw last-block DFA constructed only from the defining window condition;
3. direct enumeration of canonical restricted-growth words for small cases;
4. the first 10×8 values displayed on A242239;
5. the column recurrence, stable-row formula, A242235, A145113, and the exact polynomial identity underlying (14).

The script also generates table 1, preventing manual transcription from becoming part of the verification chain.

AI systems were used during exploration, adversarial checking, and drafting. The mathematical deliverable is the explicit proof above together with a deterministic, inspectable verification program; no AI output is treated as an oracle or as evidence by itself.

Acknowledgment

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References

[1] OEIS Foundation Inc., *A242239: Number of length $n+k+1$ arrays with every value appearing in every consecutive $k+2$ elements*, `https://oeis.org/A242239`, accessed July 23, 2026.

[2] OEIS Foundation Inc., *A242235*, `https://oeis.org/A242235`, accessed July 23, 2026.

[3] OEIS Foundation Inc., *A145113: Binary words with fewer than five zeros between consecutive ones*, `https://oeis.org/A145113`, accessed July 23, 2026.

[4] OEIS Foundation Inc., *A004767:* $a(n) = 4n+3$, `https://oeis.org/A004767`, accessed July 23, 2026.

[5] R. P. Stanley, *Enumerative Combinatorics, Volume 1*, 2nd ed., Cambridge University Press, 2012, Section 4.7.

[6] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Chapter I.